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# Multicell MIMO Communications Relying on Intelligent Reflecting Surfaces

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**Abstract** Intelligent reflecting surfaces (IRSs) constitute a disruptive wireless communication technique capable of creating a controllable propagation environment. In this paper, we propose to invoke an IRS at the cell boundary of multiple cells to assist the downlink transmission to cell-edge users, whilst mitigating the inter-cell interference, which is a crucial issue in multicell communication systems. We aim for maximizing the weighted sum rate (WSR) of all users through jointly optimizing the active precoding matrices at the base stations (BSs) and the phase shifts at the IRS subject to each BS's power constraint and unit modulus constraint. Both the BSs and the users are equipped with multiple antennas, which enhances the spectral efficiency by exploiting the spatial multiplexing gain. Due to the non-convexity of the problem, we first reformulate it into an equivalent one, which is solved by using the block coordinate descent (BCD) algorithm, where the precoding matrices and phase shifts are alternately optimized. The optimal precoding matrices can be obtained in closed form, when fixing the phase shifts. A pair of efficient algorithms are proposed for solving the phase shift optimization problem, namely the Majorization-Minimization (MM) Algorithm and the Complex Circle Manifold (CCM) Method. Both algorithms are guaranteed to converge to at least locally optimal solutions. We also extend the proposed

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